*“Everything under heaven*
*is in chaos;*
*the situation is excellent.”*



# GABMs are a mess

- How do agents connect?
- How are they prompted?
- How are they orchestrated?
- How are they activated?
- What can each agent see at each turn?
- Are all users LLMs—or hybrid?
- Base or post-trained?
- What words do we even use for these choices?
- Generate content—or only react?
- Create connections—or fixed network?
- Does the prompt / state update?
- Do in-simulation actions update memory / state?
- Real people—or synthetic personas?
- Is only one LLM used?
- Are notifications modelled?

Every choice can bias—or determine—the output.

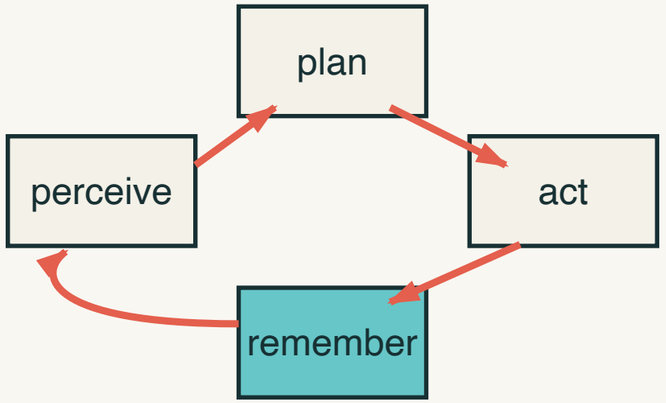
Results and claims of validity are rarely comparable.

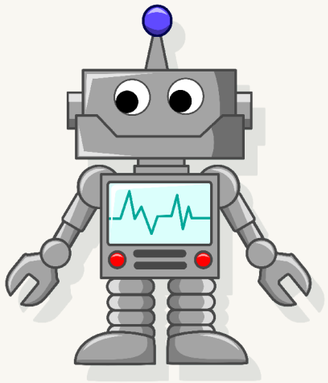
Too many degrees of freedom.

# Agentic / non-agentic

LOOP

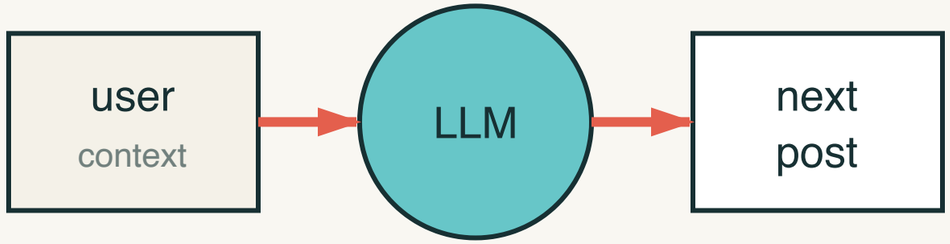
## Agentic



ONE MAPPING

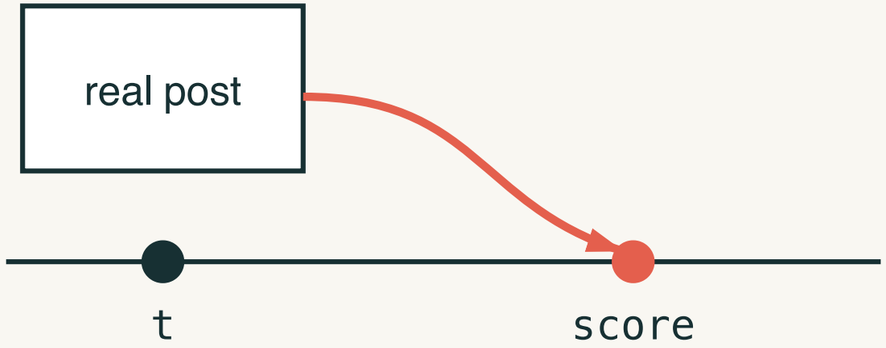
## Non-agentic



# Single-turn / multi-turn

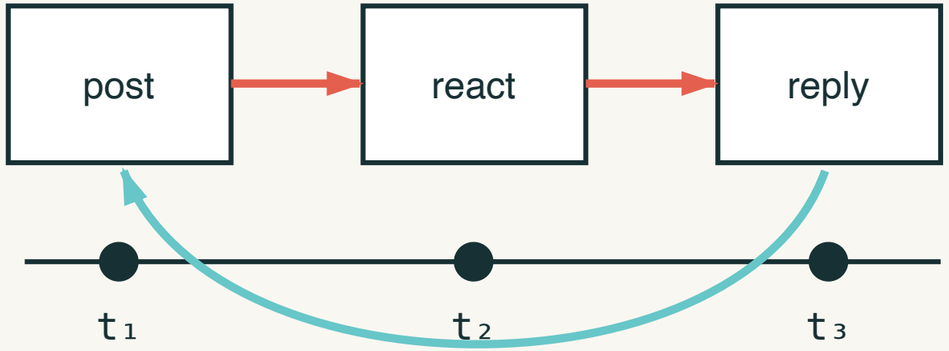
ONE OBSERVED ACTION

## Single-turn




A TRAJECTORY

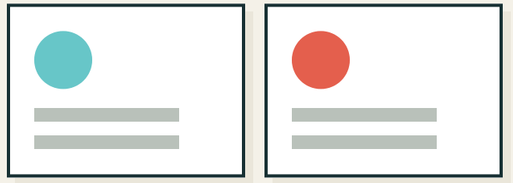
## Multi-turn



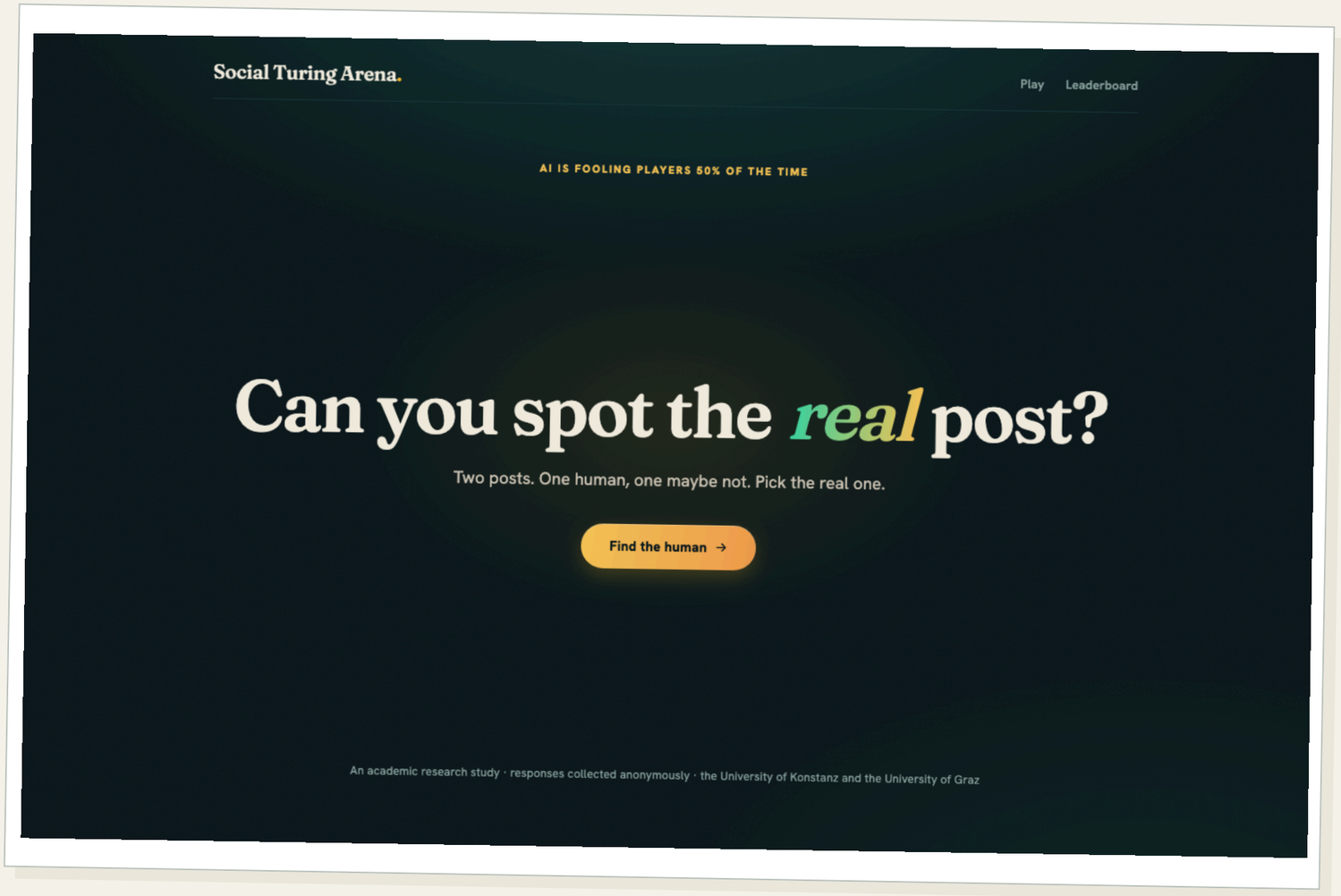

One action, or a world that feeds back?

# *Plausibility*

real / not real

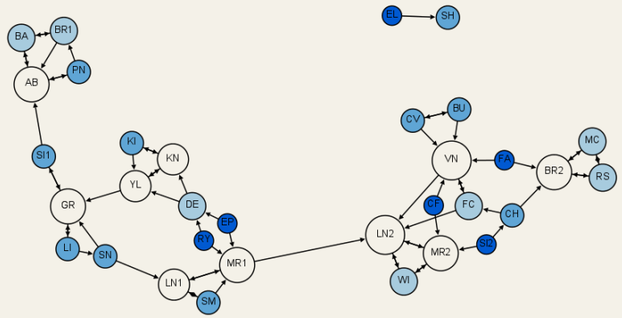


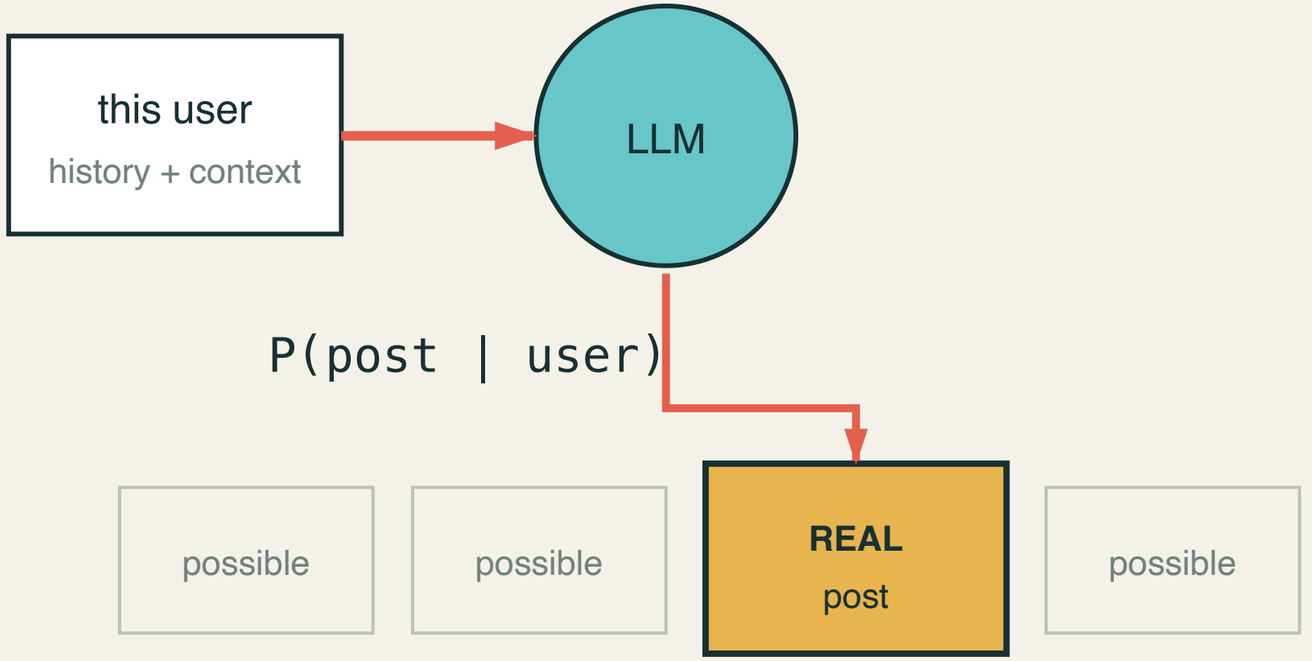
**SocialSimBench**



# Realisticness

## Persona Sweep




score what actually happened



Sociogram: Martin Grandjean, CC BY-SA 4.0 · 06 / 16

# My corner

| | NON-AGENTIC | AGENTIC |
|---|---|---|
| SINGLE-TURN | **PERSONA SWEEP** | Park et al.<br>survey agents |
| MULTI-TURN | ? | OASIS · SimUSER<br>Concordia · Y Social |

# Perplexity

$$-\log_2 P(\text{real post}) / \text{bytes}$$

**PPL → 1**

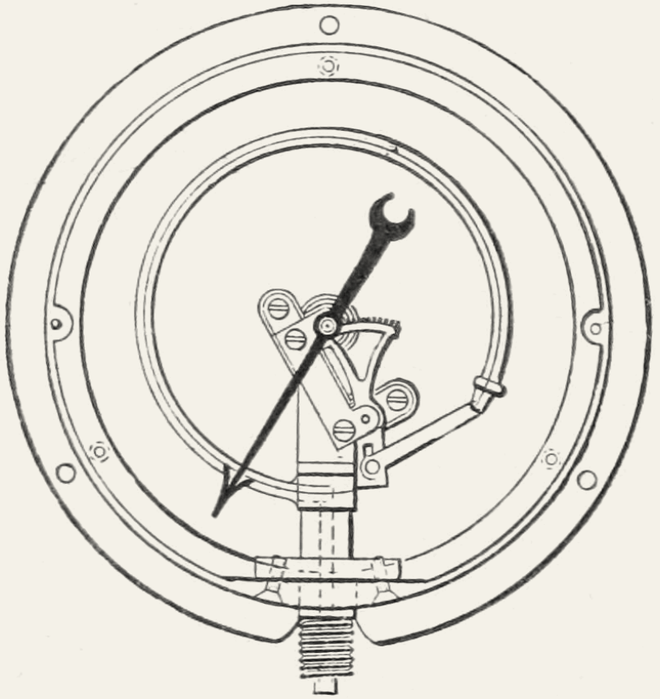
0 b/B

the real post

**PPL → ∞**

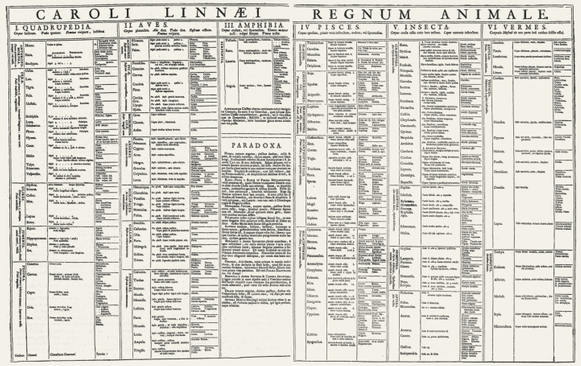
∞ b/B

no chance

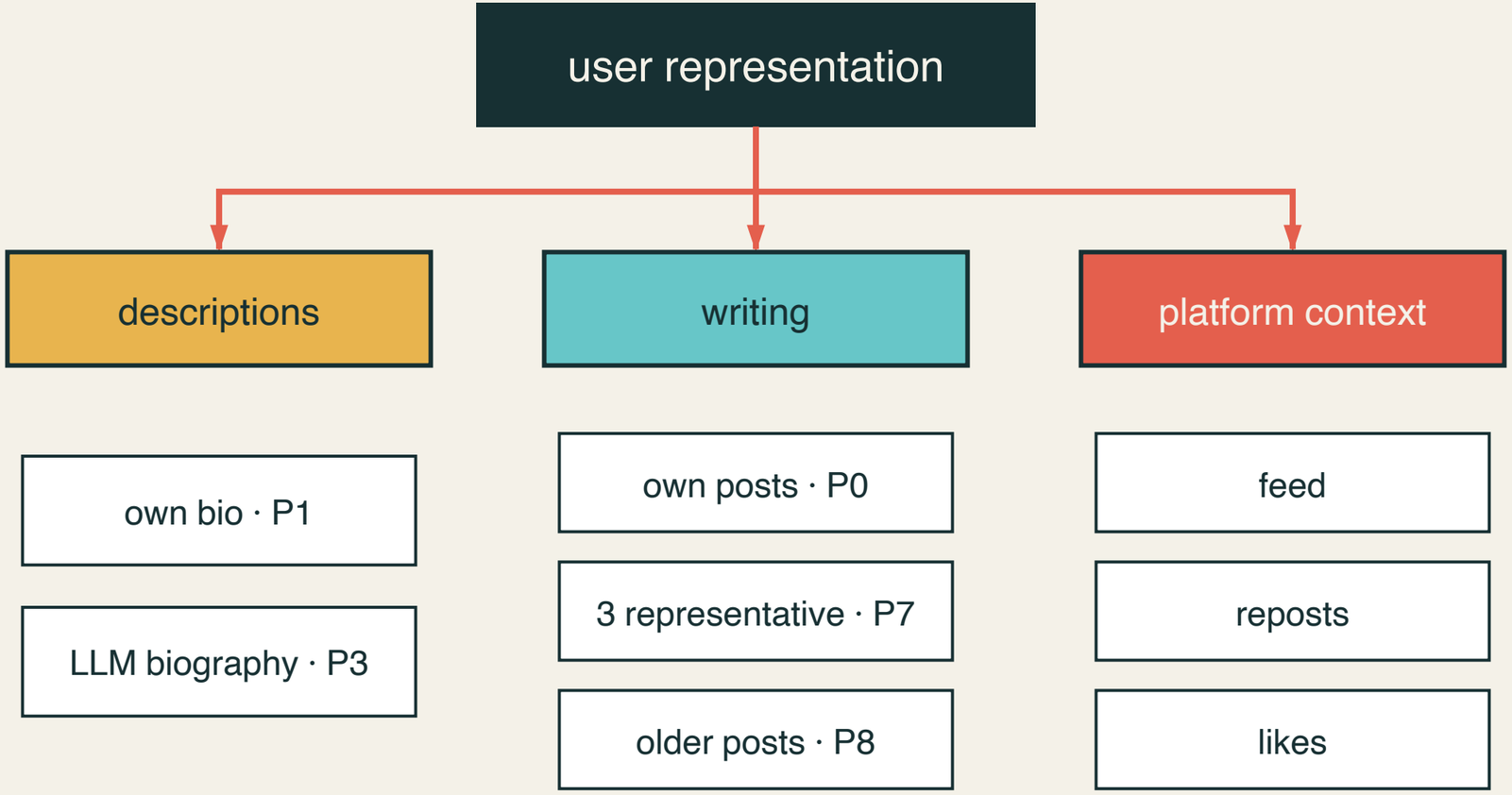


lower = better · comparable across tokenizers

# Personas

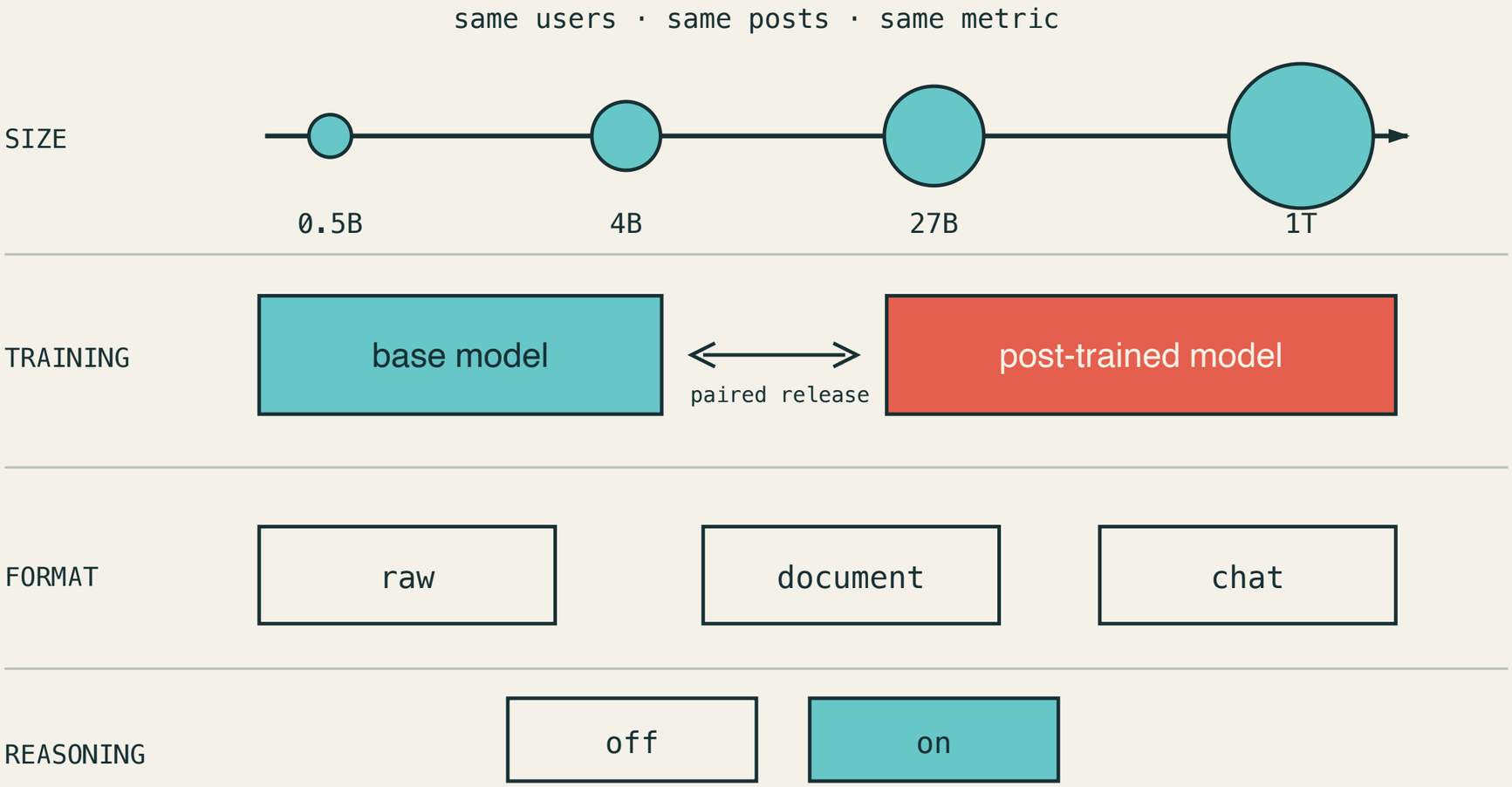

CAROLI LINNÆI REGNUM ANIMALE.
I QUADRUPEDIA
II AVES
III AMPHIBIA
IV PISCES
V INSECTA
VI VERMES
PARADOXA

What does the model know?



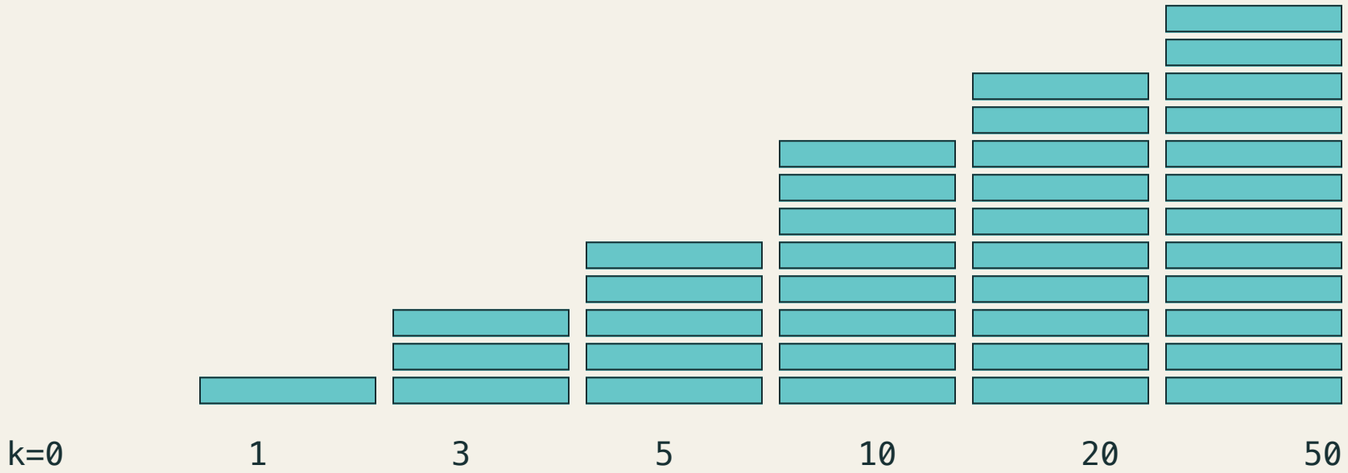

Description, demonstration, or environment?

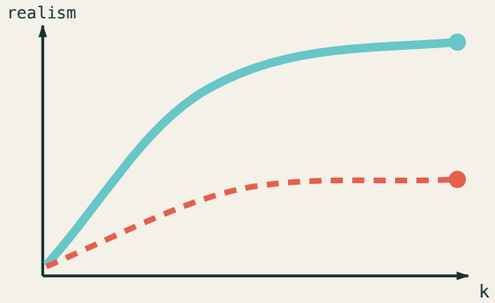
# Models




Size × training × format × reasoning.

# Dosed persona context




this user  another user (control)

# 2 platforms × 2 eras

| | BLUESKY | REDDIT |
|---|---|---|
| **OLD**<br>SEP 2023–FEB 2024 | **10,000** users<br>**50,000** posts | **10,000** users<br>**50,000** posts |
| **NEW**<br>MAR–MAY 2026 | **9,977** users<br>**49,885** posts | **10,000** users<br>**50,000** posts |

Same renderer. Same task. Four groups.

# Scale

**34**

base models

**71**

post-trained models

**18**

persona approaches

**417M**

post scores

8,619 experiments

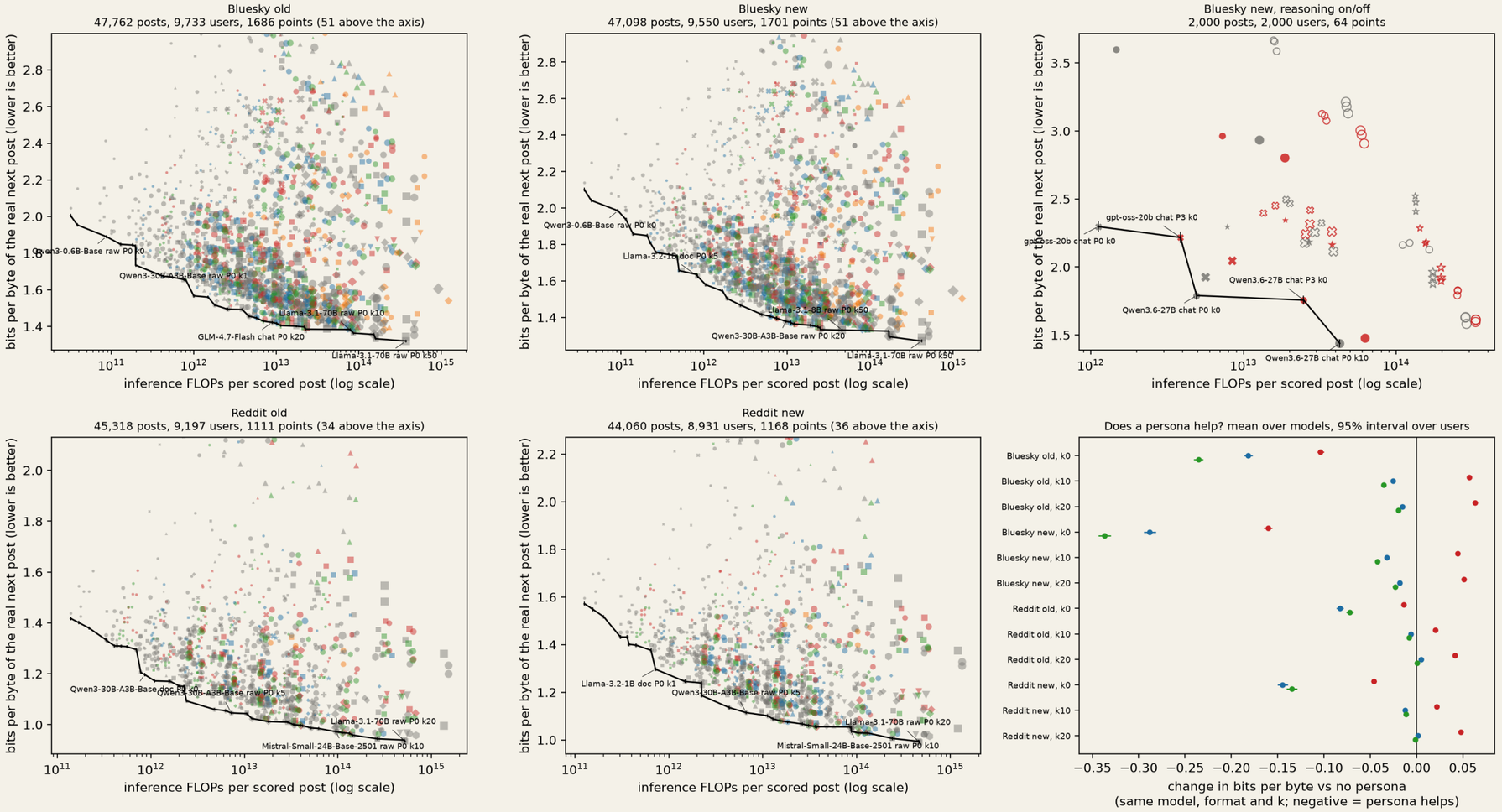
**>8,746**

GPU-hours

recorded lower bound

# Compute × context

lower = better




FLOPs per post: forward pass counted per architecture from config.json and safetensors shapes (analysis/flops_model.py): 2 x active non-embedding weights per token (MoE: router + top-k routed + shared experts), logits at every position, causal attention per query-key pair with sliding/chunked/top-k windows, linear-attention layers linear in length; each post's own length; reasoning traces decoded token by token; no prefix caching. Every point in a panel is on the same posts. Black line: Pareto frontier (lowest loss at or below each FLOPs level), bars 95% intervals over users. Reasoning panel: traces capped at 2,048 tokens, 4 samples at T0.6/T1.0 averaged per post. Old = target posts 2023-09-01..2024-01-31, new = 2026-03-01..2026-05-11, on both platforms. Bluesky old users are early adopters (first 100k accounts) with at most ~7 months of earlier posts; Reddit old users have years of earlier posts.

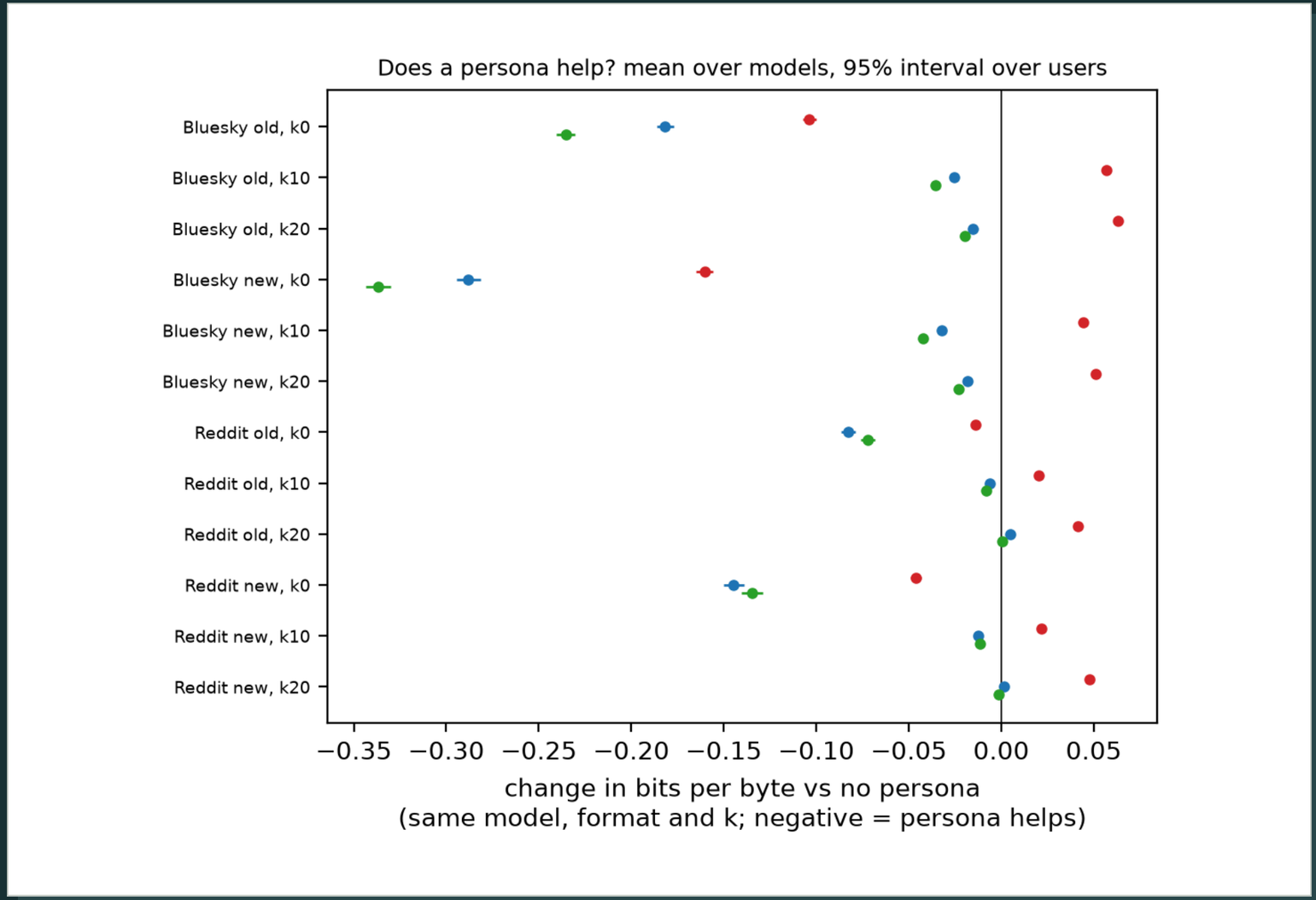
# Show user *traces.*

P8 · older own posts

P7 · 3 representative posts

P3 · LLM biography

posts > descriptions



# How do we leverage this for training?

